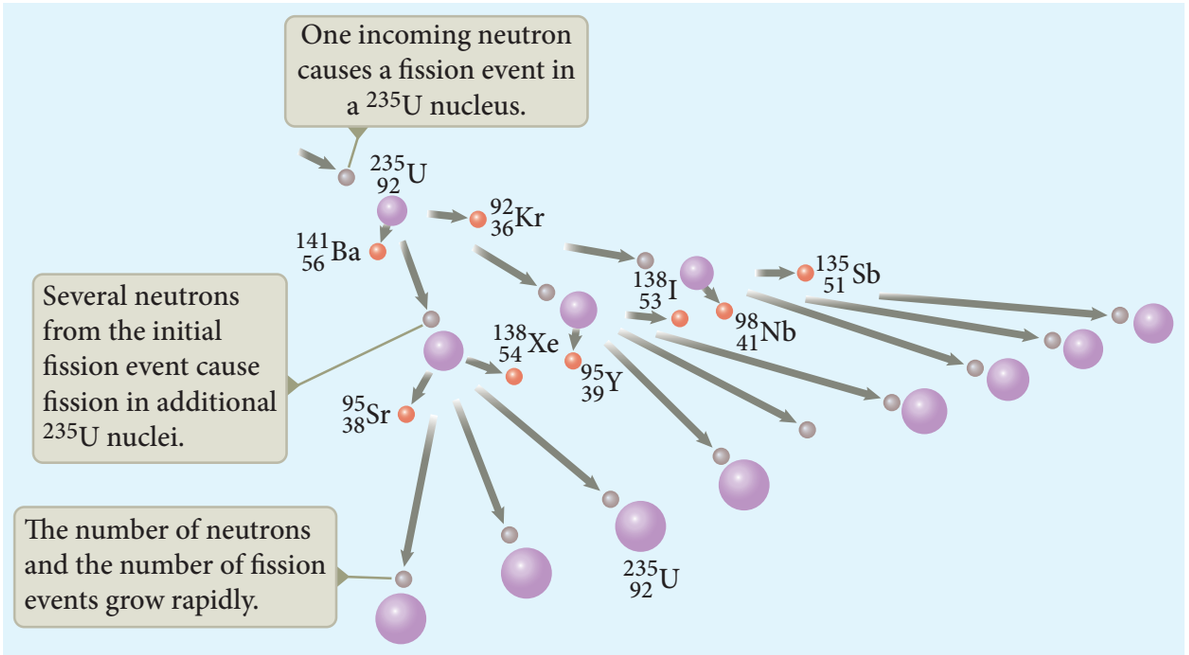


Figure 9.28 Nuclear chain reaction

one neutron initiates the fission process in one nucleus by producing three neutrons and energy of about 200 MeV. In the second step, further three nuclei undergo fission, in third step nine nuclei undergo fission, in fourth step 27 nucleus undergo fission and so on. In the 100th step, the number of nuclei which undergo fission is around 2.5×10^{40} . The total energy released after 100th step is $2.5\times10^{40}\times200MeV = 8\times10^{29}J$. It is really an enormous amount of energy which is equivalent to electrical energy required in Tamilnadu for several years.

If the chain reaction is controllable, then we can harvest an the enormous amount of energy for our needs. It is achieved in a controlled chain reaction. In the controlled chain reaction, the average number of neutrons released in each stage is kept as one such that it is possible to store the released energy. In nuclear reactors, the controlled chain reaction is carried out and the produced energy is used for power generation or for research purpose.

EXAMPLE 9.15

Calculate the amount of energy released when 1 kg of $^{235}_{92}U$ undergoes fission reaction.

Solution

235 g of $^{235}_{92}U$ has 6.02×10^{23} atoms. In one gram of $^{235}_{92}U$, the number of atoms is equal to $\frac{6.02\times10^{23}}{235} = 2.56\times10^{21}$.

So the number of atoms present in 1 kg of $^{235}_{92}U = 2.56\times10^{21}\times1000 = 2.56\times10^{24}$

Each $^{235}_{92}U$ nucleus releases 200 MeV of energy during the fission. The total energy released by 1kg of $^{235}_{92}U$ is

$$Q = 2.56\times10^{24}\times200MeV = 5.12\times10^{26}\,MeV$$

In terms of joules,

$$Q = 5.12\times10^{26}\times1.6\times10^{-13}J = 8.192\times10^{13}J\,.$$

In terms of kilowatt hour,

$$Q = \frac{8.192\times10^{13}}{3.6\times10^{6}} = 2.27\times10^{7}\,kWh$$

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